Second Welfare Theorem

Econ 2100 Fall 2018

Lecture 18, October 31 (boo)

Outline

1. Second Welfare Theorem

From Last Class

- We want to state a prove a theorem that says that any Pareto optimal allocation is (part of) a competitive equilibrium.
- That will entail finding the prices that make that allocation an equilibrium.
- First, we had to change the definition of equilibrium to deal with budget constraints.
- Given an economy $\left(\{X_i, \succsim_i, \omega_i\}_{i=1}^{I}, \{Y_j\}_{j=1}^{J}\right)$, an allocation x^*, y^* and a price vector p^* are a price equilibrium with transfers if there exists a vector of wealth levels

$$w \in \mathbb{R}^L \qquad \text{with} \qquad \sum_{i=1}^{I} w_i = p^* \cdot \sum_{i=1}^{I} \omega_i + \sum_{j=1}^{J} p^* \cdot y_j^*$$

such that:

1. For each $j = 1, ..., J$: $p^* \cdot y_j \leq p^* \cdot y_j^*$ for all $y_j \in Y_j$;
2. For each $i = 1, ..., I$: $x_i^* \succsim_i x_i$ for all $x_i \in \{x_i \in X_i : p^* \cdot x_i \leq w_i\}$;
3. $\sum_{i=1}^{I} x_i^* \leq \sum_{i=1}^{I} \omega_i + \sum_{j=1}^{J} y_j^*$, with $p_l = 0$ if the inequlity is strict for good l.

Quasi-Equilibrium

- To fix the boundary issues, we make another small definitional change.

Definition

Given an economy $\{X_i, \succsim_i\}_{i=1}^{I}, \{Y_j\}_{j=1}^{J}, \omega$, an allocation x^*, y^* and a price vector p^* constitute a quasi-equilibrium with transfers if there exists a vector of wealth levels

$$w = (w_1, w_2, ..., w_I) \quad \text{with} \quad \sum_{i=1}^{I} w_i = p^* \cdot \omega + \sum_{j=1}^{J} p^* \cdot y_j^*$$

such that:

1. For each $j = 1, ..., J$, $p^* \cdot y_j \leq p^* \cdot y_j^*$ for all $y_j \in Y_j$
2. For every $i = 1, ..., I$,
   if $x_i \succ_i x_i^*$ then $p^* \cdot x_i \geq w_i$
3. $\sum_{i=1}^{I} x_i^* = \sum_{i=1}^{I} \omega_i + \sum_{j=1}^{J} y_j^*$ with $p^* = 0$ if the inequality is strict for some i

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3. $\sum_{i=1}^{I} x_i^* \leq \sum_{i=1}^{I} \omega_i + \sum_{j=1}^{J} y_j^*$ with $p_l = 0$ if the inequlity is strict for good l.

- Make sure you see why this deals with the previous problem.
- Any equilibrium with transfers is a quasi-equilibrium (why?).

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Second Welfare Theorem: Preliminaries

- We have seen a few counterexamples to a possible second welfare theorem, and ways in which we can deal with these.
- This is a summary of what we have learned so far.

To show that for any Pareto optimal allocation one can find prices that make it into a competitive equilibrium requires a few assumptions

- We need transfers to overcome the limitations imposed by private ownership.
- We need convexity of production sets.
- We need convexity and local non-satiation of preferences.
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Theorem (Second Fundamental Theorem of Welfare Economics)

Consider an economy $\left(\{X_i, \succsim_i\}_{i=1}^{I}, \{Y_j\}_{j=1}^{J}, \omega\right)$ *and assume that* Y_j *is convex for all* $j = 1, ..., J$, *and* $\succsim_i$ *is convex and locally non-satiated for all* $i = 1, ..., I$.

Then, for each Pareto optimal allocation $\hat{x}, \hat{y}$ *there exists a price vector* $\hat{p} \neq 0$ *such that* $(\hat{x}, \hat{y}, \hat{p})$ *form a quasi-equilibrium with transfers.*

- The proof uses the separating hyperplane theorem.
- The proof is in three parts: aggregation, separation, and decentralization.
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  - If an allocation is Pareto optimal there is an hyperplane that simultaneously supports the better-than sets of all consumers and all producers.
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Proof of the Second Welfare Theorem: Aggregation

- First, we aggregate all consumers preferences when evaluating the Pareto efficient consumption bundle $\hat{x}$.
- Define the following sets:

$$V_i = \{x_i \in X_i : x_i \succ \hat{x}_i\} \subset \mathbb{R}^L \quad \text{and} \quad V = \sum_i V_i$$

- **Claim**: *V is convex.*
  - *Take $x, x' \in V$ (so there are strictly preferred consumption bundles x_i for x and x'_i for x').*
  - *Since preferences are convex, for any $\lambda \in [0,1]$*

$$\lambda x_i + (1-\lambda) x'_i \succ \hat{x}_i$$

  - *Therefore, $\lambda x + (1-\lambda) x' \in V$ (a strictly preferred bundle to $\hat{x}$).*

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  - V is the set of all bundles strictly preferred to $\hat{x}$ by every consumer.

- **Claim**: *V is convex.*
  - *Take $x, y \in V$ (so that x is strictly preferred to $\hat{x}$ by every consumer i, and y is strictly preferred to $\hat{x}$ by every consumer i).*
  - *Since preferences are convex, for each $\lambda \in [0,1]$, $\lambda x + (1-\lambda)y$ is weakly preferred to $\hat{x}$ by every consumer i.*
  - *Moreover, $\lambda x + (1-\lambda)y$ is strictly preferred to $\hat{x}$ by every consumer i for $\lambda \in (0,1)$.*

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  - *Since preferences are convex, for each $\lambda \in [0, 1]$,*
  - *By transitivity, we have*
  - *Therefore, $\lambda x + (1 - \lambda) y \in V$, so V is convex.*

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$$\lambda x' + (1-\lambda)\, x'' \succsim_i x'' \succ_i \hat{x}_i$$
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  - *V is convex because it is the sum of I convex sets.*

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  - *Therefore, $\lambda x' + (1-\lambda)\, x''$ is an element of V_i and therefore each V_i is convex.*
  - *V is convex because it is the sum of I convex sets.*

Proof of the Second Welfare Theorem: Aggregation

- Second, we aggregate all firms and define the set of attainable consumption bundles.

- Define the aggregate production set as

$$Y = \sum_j Y_j = \left\{ \sum_j y_j \in \mathbb{R}^L : y_1 \in Y_1, ..., y_J \in Y_J \right\}$$

- The set of consumption bundles that can be allocated to consumers is

$$Y + \{\omega\}$$

- This set is convex since it is the sum of $J + 1$ convex sets.

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- Draw V and $Y + \{\omega\}$.

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Proof of the Second Welfare Theorem: Separation

- Next, we separate the sets V and $Y + \{\omega\}$.
- Since $(\hat{x}, \hat{y})$ is a Pareto optimal allocation, $V \cap Y + \{\omega\} = \emptyset$.
- Since V and $Y + \{\omega\}$ are two disjoint convex sets, one can apply the Separating Hyperplane Theorem.

Separate V and $Y + \{\omega\}$

By the Separating Hyperplane Theorem, there exist a $\hat{p} \in \mathbb{R}^L$ with $\hat{p} \neq 0$ and an $r \in \mathbb{R}$ such that

$$\hat{p} \cdot z \geq r \quad \text{for all } z \in V,$$

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- Next, we look at the implication of separation for consumers. By separation,

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Claim: *if $x_i \succsim_i \hat{x}_i$ for all i, then $\hat{p} \cdot (\sum_i x_i) \geq r$*

remember this as we will use it later

- *Take any $x_i \succsim_i \hat{x}_i$ for all i.*
- *By local nonsatiation, for each i there exists x_i' preferred to x_i that is arbitrarily close to x_i.*
- *Hence, $x_i' \succ_i \hat{x}_i$ for all i, and $\sum_i x_i' \in V$.*
- *So, by (2), $\hat{p} \cdot (\sum_i x_i') \geq r$ (by separation).*
- *Take a sequence of x_i' that goes to x_i and use the fact that the budget set is closed.*

- Applying this result to $\hat{x}_i \succsim_i \hat{x}_i$, separation tells us that

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- We have shown that $\sum_i \{x_i \in X_i : x_i \succsim_i \hat{x}_i\}$ belongs to the closure of V which is contained in the half-space $\left\{ z \in \mathbb{R}^L : \hat{p} \cdot z \geq r \right\}$.

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- *So,* $\hat{p} \cdot (\sum_i \check{x}_i) \geq r$ *(by separation);*
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- The Pareto optimal allocation is feasible, and therefore:
$$\sum_i \hat{x}_i \leq \sum_j \hat{y}_j + \omega \in Y + \{\omega\}$$
- Hence we have
$$\hat{p} \cdot \left(\sum_i \hat{x}_i \right) \leq r$$
- Putting together this inequality and the opposite one from the previous slide:
$$\hat{p} \cdot \left(\sum_i \hat{x}_i \right) = r$$

Geometry of the Proof

- $\sum_i \hat{x}_i$ belongs to $Y + \{\omega\}$ and it lies in the half-space $\left\{ z \in \mathbb{R}^L : \hat{p} \cdot z \leq r \right\}$.

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- *As shown previously,*

$$\hat{p} \cdot \underbrace{\left(x + \sum_{n \neq i} \hat{x}_n \right)}_{\text{this satisfies } x_i \gtrsim_i \hat{x}_i \text{ for all } i} \geq r$$

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- *Set $w_i = \hat{p} \cdot \hat{x}_i$ so that we have $\hat{p} \cdot x \geq w_i$ as desired.*

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$$y_j + \sum_{k \neq j} \hat{y}_k \in Y$$

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$$\hat{p} \cdot \left(\omega + y_j + \sum_{k \neq j} \hat{y}_k \right) \leq \omega + \hat{p} \cdot \left(\omega + \hat{y}_j + \sum_{k \neq j} \hat{y}_k \right)$$

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- *The above inequality holds for any $y_j \in Y_j$*

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  - Then, add strict monotonicity, and show that a quasi equilibrium with transfers which has strictly positive wealth for all consumers is an equilibrium with transfers.

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What Are The Welfare Theorems About?

- The first welfare theorem says a competitive equilibrium is Pareto efficient: markets can yield efficient allocations.
- The second welfare theorem says that any Pareto efficient allocation can be obtained as an equilibrium provided one makes the 'right' adjustment to income.
- In other words, any outcome that maximizes social welfare can be obtained by redistributing income 'correctly' across consumers.
- There are many caveats to these results
  - Both theorems rule out externalities.
  - Both theorems need local non satiation.
  - For the second welfare theorem, even assuming all assumptions are satisfied, we need someone to decide what are the transfers.
    - This is not practical as this someone would need to know everyone's preferences, and all production sets, to figure those transfers out. How do you know those?
    - In public economics, and optimal taxation theory, one asks if there is a way to get the consumers to reveal their preferences to the planner.

Next Week

- Equilibrium characterization in the differentiable case.
- Welfare theorems in the differentiable case.
- Existence.